CUSTOMER ANALYTICS / PROJECT PRESENTATION

Feature Engineering with Python

Customer Churn Analysis using Churn_Modelling.csv

Python • pandas • NumPy • Seaborn • scikit-learn

Prepared by: [Your Name]

Project overview

Business problem

Customer churn occurs when a customer stops doing business with an organization. In banking, churn can affect revenue, customer lifetime value, and growth.

Project objectives

Inspect and clean customer data

Create meaningful features from existing columns

Explore patterns associated with churn

Prepare data for classification models

Target variable: Exited

1 = churned | 0 = retained

Dataset and tools

Typical dataset fields

Demographics

Age, Gender, Geography

Bank relationship

Tenure, NumOfProducts, IsActiveMember

Financial indicators

CreditScore, Balance, EstimatedSalary

Outcome

Exited

Technology stack

pandas — data manipulation

NumPy — numerical operations

Matplotlib / Seaborn — charts

scikit-learn — preprocessing and modeling

Note: column names and available fields can vary by dataset version.

Data preparation

01 Load

Read CSV and inspect rows, columns, types, and summary statistics.

02 Audit quality

Check missing values, duplicates, unexpected categories, and ranges.

03 Clean

Resolve data issues using documented, appropriate rules.

04 Remove identifiers

Exclude RowNumber, CustomerId, and Surname from model predictors when present.

Feature engineering

AgeGroup

Convert age into bands

Compare churn across age segments.

TenureGroup

Group years with the bank

Explore retention by relationship length.

HasBalance

1 if Balance > 0

Separate zero-balance and positive-balance customers.

MultipleProducts

1 if NumOfProducts > 1

Explore whether product breadth relates to churn.

AgeTenureInteraction

Age × Tenure

Allow a model to test a combined relationship.

Categorical encoding

One-hot encode categories

Represent Geography and Gender for many models.

Python implementation


```
import pandas as pd

df = pd.read_csv("Churn_Modelling.csv")

df = df.drop(
    columns=["RowNumber", "CustomerId", "Surname"],
    errors="ignore"
)

df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[0, 25, 35, 45, 55, 100],
    labels=["18-25", "26-35", "36-45", "46-55", "56+"]
)
df["HasBalance"] = (df["Balance"] > 0).astype(int)
df["MultipleProducts"] = (df["NumOfProducts"] > 1).astype(int)
df["AgeTenureInteraction"] = df["Age"] * df["Tenure"]

print(df.groupby("AgeGroup", observed=True)["Exited"].mean())
```

Code notes

Confirm column names match your file.

Inspect missing values before modeling.

Treat engineered features as hypotheses to test.

Fit learned preprocessing on training data only.

Exploratory data analysis

Questions to answer with the data

How does churn vary by age group?

Are churn rates different across geographies?

Does tenure relate to customer retention?

How does churn vary by number of products?

Is customer activity associated with churn?

Recommended visualizations

Bar chart: churn rate by AgeGroup

Bar chart: churn rate by Geography

Stacked chart: churn by product count

Heatmap: numeric feature correlations

Report actual rates and sample sizes; do not assume which segment churns most.

Modeling and evaluation plan

Baseline model

Logistic Regression provides an interpretable starting point.

Tree-based model

Random Forest can capture nonlinear patterns and interactions.

Compare fairly

Use the same train/test split and consistent evaluation.

Evaluation metrics

Precision Of predicted churners, how many actually churned?

Recall Of actual churners, how many were identified?

F1-score Balances precision and recall.

ROC-AUC Measures ranking ability across thresholds.

Results and interpretation

Complete this slide after running the analysis

Dataset size	[Add rows × columns]
Overall churn rate	[Add calculated percentage]
Highest churn segment	[Add evidence-backed segment]
Baseline model F1 / ROC-AUC	[Add measured scores]
Engineered model F1 / ROC-AUC	[Add measured scores]

No performance scores or dataset findings have been invented; replace placeholders with your outputs.

Limitations and good practices

Avoid data leakage

Split data before fitting imputers, encoders, scalers, or selecting features.

Validate engineered features

Compare performance with and without each feature using a fair experiment.

Check class imbalance

Review class distribution and use appropriate metrics and thresholds.

Interpret responsibly

Association does not prove causation; consider privacy and fairness.

The best features depend on data quality, modeling choices, and the business context.

CONCLUSION

Turn customer data into retention insights



Feature engineering can make customer characteristics easier to analyze.

EDA identifies segments and relationships worth investigating.

Model comparison tests whether engineered features add predictive value.

Evidence-based findings can inform customer retention experiments.

Thank you

Questions?